

Introduction

Contributors

Team members



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Interests:

Roles

Client

Project Definition

Research and Subquestions

The main research question of this project is:

How to create a harmonised cross-border DEM (of the Rhine catchment area) in an automated (and efficient) way?

As described in the introduction, the project focusses on the issues presented by the cross-border data of the Rhine catchment area. This translates to the creation of an automated pipeline and the report describing the issues that come with attempting the creation of a harmonised DEM. The efficiency refers to creating a pipeline that accounts for the large data quantity and limited computational resources. These two aspects of the main question can be broken down into the following subquestions:

- What are the limitations of an automated harmonisation pipeline?
- What are the unknowns/variables in the data we need to consider?
- What are the existing DEMs (local/global)?
- What, if any, are the existing workflows for processing point clouds from heterogeneous sources?

Answering these subquestions will help to answer the main research question. The first two subquestions are more theoretical and will be answered in the report. The last two subquestions are more practical and will be answered in the pipeline.

Relevant courses from the MSc Geomatics program

Requirements

Req ID	Description	MoSCoW
DT-01	Global EU DEM (as ground truth)	Must
DT-02	Rhine Watershed Mask	Must
DT-03	Rhine Bathymetry	Could
CT-01	Netherlands is included	Must
CT-02	Germany is included	Must
CT-03	Belgium is included	Should
CT-04	Switzerland is included	Must
CT-05	France is included	Must
CT-06	Luxembourg is included	Must
CT-07	Austria is included	Must
CT-08	Liechtenstein is included (included in Swiss data)	Must
CT-09	Italy is included	Won't have
MP-01	DSM map	Must
MP-02	DTM map	Must
MP-03	Land-sea mask	Must
MP-04	Bathymetry mask	Could
MP-05	Nodata mask	Should
MP-06	Point cloud density map	Should
TC-01	Workflow is fully automated	Must
TC-02	Workflow is efficient	Should
TC-03	Workflow is reproducible	Must
RP-01	Report on findings	Must
RP-02	Report details the workflow	Must
RP-03	Report details the issues with cross-border data	Must
RP-04	Report details the limitations of the workflow	Must

Table 1: MoSCoW prioritization of data requirements for the project

Methodology

Research Approach

Research Methodology

Quality Assurance

Planning

Phases

Communication

Communication is key to the success of this project. To ensure communication is maintained, both within the team and with stakeholders, regular meetings will be held. The team will meet on a weekly basis to discuss progress, issues and questions. These meetings will be held on Fridays. There will be an additional opportunity to meet on Mondays if necessary. For each meeting, key talking points will be shared in advance. Meeting minutes will be kept and distributed after each meeting. These meetings will be held online via Microsoft Teams. The team will also communicate internally outside of these meetings, either through Whatsapp or in person.

Risk Analysis

Risk ID	Description	Impact	Likelihood	Mitigation
1.	Pipeline too computationally taxing or insufficient computational resources.	Critical	Possible	Test with small dataset and adjust spatial extent if necessary.
2.	Insufficient quality/availability of pointcloud data.	Marginal	Rare	Shift focus to regions with available high-quality data.
3.	CRS transformations are inaccurate.	Critical	Unlikely	TODO: Test CRS alignment on select border regions
4.	Missing/conflicting data on border areas.	Critical	Likely	Check spatial overlap of datasets and prioritize one dataset.
5.	Task is too ambitious given timeframe/team size.	Unlikely	Catastrophic	Check if internal deadlines are met and adjust scope if necessary.
6.	Edge cases not properly assessed due to data quantity and variability.	Minor	Possible	Unideal but acceptable within scope of project.
7.	Task is insufficiently constrained/defined.	Critical	Unlikely	Regular meetings with team and stakeholders to assess and adjust scope and requirements if necessary.
8.	Lacking/inadequate communication within the team and/or with stakeholders.	Critical	Possible	Schedule regular meetings and maintain open communication channels.
9.	Discovery of unexpected issues.	Marginal	Possible	Regular meetings with team and stakeholders to assess impact and devise mitigation strategies if necessary.
10.	Temporary or permanent absence of team members.	Critical	Rare	Set internal deadlines early to allow for adjustments in case of absence.
11.	Data loss.	Catastrophic	Possible	Implement regular backup procedures.

Table 2: Risk Assessment Table.

Likelihood	Harm severity			
	Minor	Marginal	Critical	Catastrophic
Certain	High	High	Very high	Very high
Likely	Medium	High	High	Very high
Possible	Low	Medium	High	Very high
Unlikely	Low	Medium	Medium	High
Rare	Low	Low	Medium	Medium
Eliminated	Eliminated			

Figure 1: Risk Assessment Matrix.

https://en.wikipedia.org/wiki/Risk_matrix